

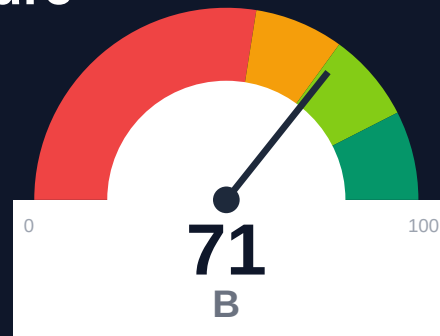
Swell

Documentation Infrastructure Audit Report

Prepared by VeriOps

2026-04-29 10:58 UTC

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Risk: Moderate

Key Metrics

Pages scanned: 331

Report type: Premium Executive Audit

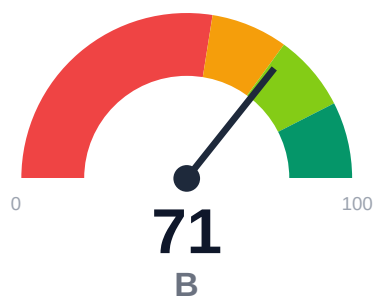
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 10
- SEO/GEO issue rate is high: 24.62%
- API usage-doc coverage is low: 31.25% (279 API pages detected)

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Documentation audit of developers.swell.is covered 331 pages with 100% crawl coverage and 91.5% overall confidence. The site contains 10 confirmed broken internal links requiring immediate remediation. API reference coverage is critically low at 31.25% despite 279 API pages detected, indicating significant gaps in usage documentation. SEO/geo issues affect 24.62% of pages, impacting discoverability. Last-updated metadata is present on only 56.5% of pages, limiting transparency on content freshness.

External posture: SEO/GEO issue rate **24.6%**, public API coverage **31.2%**, broken links **10**.

71 Audit Score	B Grade	Moderate Risk Band
331 Pages Scanned	10 Broken Links	24.6% SEO Issue Rate

Per-Site Breakdown

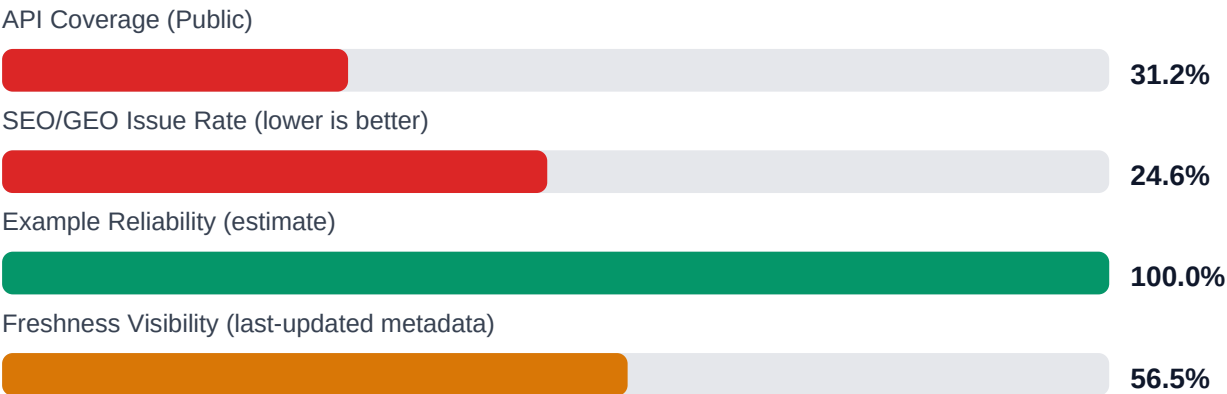
Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://developers.swell.is/	331	10	31.2	100.0	24.6

Industry Benchmark Comparison

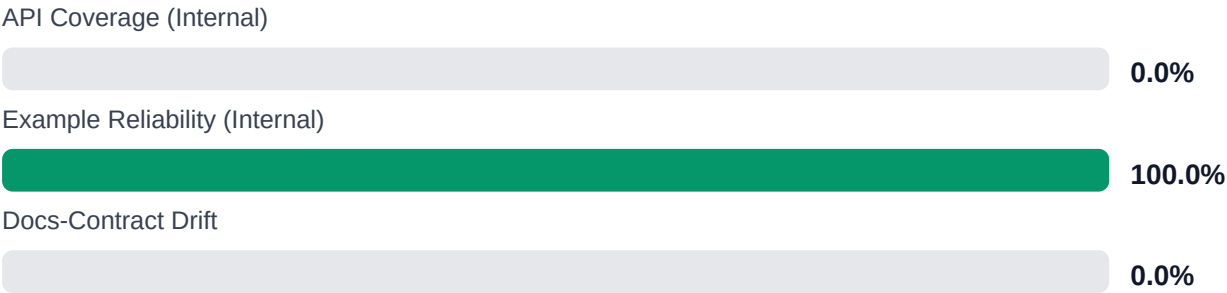
Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-21
Industry average (developer docs)	85	-14
Minimum acceptable	70	+1
Your score	71	-

Gap to industry average: **14 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics



Internal Metrics (from scorecard)



Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
\$140,822 annual exposure Remediation: \$2,090 Monthly loss: \$4,125 Opportunity: \$7,436/mo	\$249,513 annual exposure Remediation: \$5,445 Monthly loss: \$12,903 Opportunity: \$7,436/mo	\$431,728 annual exposure Remediation: \$8,800 Monthly loss: \$27,808 Opportunity: \$7,436/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	\$1,782	\$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$1,122	\$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	\$2,257	\$990
F-API-COVERAGE	Undocumented API operations	LOW	\$2,904	\$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$2,244	\$935
F-DRIFT	Code/docs contract drift	LOW	\$1,901	\$660
F-TERMINOLOGY	Terminology inconsistency	LOW	\$693	\$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

1. Fix all 10 confirmed broken internal links across documentation [impact: critical, effort: low]
2. Audit 279 API pages and create usage documentation to raise coverage from 31.25% to minimum 70% [impact: critical, effort: high]

3. Remediate SEO/geo issues on 81 affected pages (24.62% of site) [impact: high, effort: medium]
4. Add last-updated timestamps to 144 pages currently missing metadata (43.5%) [impact: medium, effort: low]
5. Implement automated link checking in CI/CD to prevent future broken link accumulation [impact: high, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

- + Complete crawl coverage achieved (331/331 pages, 100%)
- + High overall audit confidence at 91.5%
- + Example code reliability estimated at 100%
- + No repository navigation link breakage detected
- + Crawl trap detection prevented 9 URLs from skewing results

Risks

- 10 confirmed broken internal links degrade user experience and SEO authority
- API reference coverage at 31.25% leaves 68.75% of API surface undocumented for usage patterns
- SEO/geo issues on 24.62% of pages (approximately 81 pages) harm search visibility
- 43.5% of pages lack last-updated metadata, eroding user trust in content currency
- 279 API pages detected but low coverage suggests incomplete integration examples or tutorials
- Single-product topology may hide cross-product documentation gaps if ecosystem expands
- API confidence at 85% indicates potential detection limitations in coverage calculation

Limitations

- * External audit cannot verify accuracy or completeness of technical content, code examples, or API specifications
- * API coverage detection at 85% confidence may undercount or misclassify reference vs. usage documentation
- * Crawl cannot assess gated content, authentication-required pages, or dynamically generated documentation
- * SEO/geo issue detection does not guarantee search engine ranking outcomes or international accessibility compliance
- * Example reliability estimate cannot execute code or validate against live API endpoints

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.